

## 10.5 Conditioning and perturbation bounds

Sections 10.2 to 10.4 were about algorithms. This section is about the problem itself: if the data  $(\mathbf{X}, \mathbf{y})$  change a little, how much can the least squares solution and the residuals change? The answer does not depend on the algorithm. Combined with backward stability through Eq. 10.1.2, it predicts the accuracy of any computed solution.

### 10.5.1 The perturbation theorem

Throughout this section  $\mathbf{X}$  is  $n \times p$  with full column rank,  $\hat{\boldsymbol{\beta}} = \mathbf{X}^+ \mathbf{y}$  is its least squares solution (assumed nonzero) and  $\hat{\boldsymbol{\epsilon}} = \mathbf{y} - \mathbf{X}\hat{\boldsymbol{\beta}}$  its residual. Write  $\kappa = \kappa(\mathbf{X}) = \sigma_1/\sigma_p$ . The perturbed data are  $\tilde{\mathbf{X}} = \mathbf{X} + \mathbf{E}$  and  $\tilde{\mathbf{y}} = \mathbf{y} + \mathbf{f}$ , with

$$\|\mathbf{E}\|_2 \leq \epsilon \|\mathbf{X}\|_2, \quad \|\mathbf{f}\| \leq \epsilon \|\mathbf{y}\|. \quad (10.5.1)$$

The solution and residual for the perturbed data are  $\tilde{\mathbf{b}}$  and  $\tilde{\boldsymbol{\epsilon}}$ . A single number measures how badly the model fits relative to the size of the problem:

$$\eta = \frac{\|\hat{\boldsymbol{\epsilon}}\|}{\|\mathbf{X}\|_2 \|\hat{\boldsymbol{\beta}}\|}.$$

Since  $\|\mathbf{X}\hat{\boldsymbol{\beta}}\| \leq \|\mathbf{X}\|_2 \|\hat{\boldsymbol{\beta}}\|$ ,  $\eta \leq \tan \theta$ , where  $\theta$  is the angle between  $\mathbf{y}$  and  $\mathcal{C}(\mathbf{X})$ . A model that fits well has small  $\eta$ .

#### Theorem 10.5.1 — Perturbation of least squares

Under Eq. 10.5.1 with  $\kappa\epsilon < 1$ , the matrix  $\tilde{\mathbf{X}}$  has full column rank, and

$$\frac{\|\tilde{\mathbf{b}} - \hat{\boldsymbol{\beta}}\|}{\|\hat{\boldsymbol{\beta}}\|} \leq \frac{\kappa\epsilon}{1 - \kappa\epsilon} (2 + \eta) + \frac{\kappa^2\epsilon}{(1 - \kappa\epsilon)^2} \eta, \quad (10.5.2)$$

$$\frac{\|\tilde{\boldsymbol{\epsilon}} - \hat{\boldsymbol{\epsilon}}\|}{\|\mathbf{y}\|} \leq \epsilon \left( 1 + \frac{2\kappa}{1 - \kappa\epsilon} \right). \quad (10.5.3)$$

To first order in  $\epsilon$ , the coefficient bound is  $\epsilon(2\kappa + \kappa(\kappa + 1)\eta)$ .

*Proof.* By Proposition 10.4.2,  $\sigma_p(\tilde{\mathbf{X}}) \geq \sigma_p - \|\mathbf{E}\|_2 \geq \sigma_p(1 - \kappa\epsilon) > 0$ , so  $\tilde{\mathbf{X}}$  has full column rank and  $\tilde{\mathbf{X}}^+ \tilde{\mathbf{X}} = \mathbf{I}$ . Hence

$$\tilde{\mathbf{b}} - \hat{\boldsymbol{\beta}} = \tilde{\mathbf{X}}^+ \tilde{\mathbf{y}} - \tilde{\mathbf{X}}^+ \tilde{\mathbf{X}} \hat{\boldsymbol{\beta}} = \tilde{\mathbf{X}}^+ (\tilde{\mathbf{y}} - \tilde{\mathbf{X}} \hat{\boldsymbol{\beta}}) = \tilde{\mathbf{X}}^+ (\hat{\boldsymbol{\epsilon}} + \mathbf{f} - \mathbf{E} \hat{\boldsymbol{\beta}}).$$

The residual term simplifies because  $\mathbf{X}^\top \hat{\boldsymbol{\epsilon}} = \mathbf{0}$ :  $\tilde{\mathbf{X}}^+ \hat{\boldsymbol{\epsilon}} = (\tilde{\mathbf{X}}^\top \tilde{\mathbf{X}})^{-1} (\tilde{\mathbf{X}}^\top \mathbf{X} + \mathbf{E}^\top) \hat{\boldsymbol{\epsilon}} = (\tilde{\mathbf{X}}^\top \tilde{\mathbf{X}})^{-1} \mathbf{E}^\top \hat{\boldsymbol{\epsilon}}$ . So

$$\tilde{\mathbf{b}} - \hat{\boldsymbol{\beta}} = \tilde{\mathbf{X}}^+ (\mathbf{f} - \mathbf{E} \hat{\boldsymbol{\beta}}) + (\tilde{\mathbf{X}}^\top \tilde{\mathbf{X}})^{-1} \mathbf{E}^\top \hat{\boldsymbol{\epsilon}}. \quad (10.5.4)$$

Now  $\|\tilde{\mathbf{X}}^+\|_2 = 1/\sigma_p(\tilde{\mathbf{X}})$  and  $\|(\tilde{\mathbf{X}}^\top \tilde{\mathbf{X}})^{-1}\|_2 = 1/\sigma_p(\tilde{\mathbf{X}})^2$  (Theorem 1.8.1). Also  $\|\mathbf{y}\| \leq \|\mathbf{X}\hat{\boldsymbol{\beta}}\| + \|\hat{\boldsymbol{\epsilon}}\| \leq \sigma_1\|\hat{\boldsymbol{\beta}}\| + \|\hat{\boldsymbol{\epsilon}}\|$ , so  $\|\mathbf{f} - \mathbf{E}\hat{\boldsymbol{\beta}}\| \leq \epsilon(\|\mathbf{y}\| + \sigma_1\|\hat{\boldsymbol{\beta}}\|) \leq \epsilon\sigma_1\|\hat{\boldsymbol{\beta}}\|(2 + \eta)$ . Therefore

$$\|\tilde{\mathbf{b}} - \hat{\boldsymbol{\beta}}\| \leq \frac{\epsilon\sigma_1\|\hat{\boldsymbol{\beta}}\|(2 + \eta)}{\sigma_p(1 - \kappa\epsilon)} + \frac{\epsilon\sigma_1\|\hat{\boldsymbol{\epsilon}}\|}{\sigma_p^2(1 - \kappa\epsilon)^2}.$$

Dividing by  $\|\hat{\boldsymbol{\beta}}\|$  and using  $\|\hat{\boldsymbol{\epsilon}}\| = \eta\sigma_1\|\hat{\boldsymbol{\beta}}\|$  gives Eq. 10.5.2.

For the residual,  $\tilde{\boldsymbol{\epsilon}} = (\mathbf{I} - \tilde{\mathbf{M}})\tilde{\mathbf{y}}$ , where  $\tilde{\mathbf{M}}$  projects onto  $\mathcal{C}(\tilde{\mathbf{X}})$ . Write  $\tilde{\mathbf{y}} = \tilde{\mathbf{X}}\hat{\boldsymbol{\beta}} + (\mathbf{f} - \mathbf{E}\hat{\boldsymbol{\beta}}) + \hat{\boldsymbol{\epsilon}}$ . The first term is annihilated by  $\mathbf{I} - \tilde{\mathbf{M}}$ , and  $\tilde{\mathbf{M}}\hat{\boldsymbol{\epsilon}} = (\tilde{\mathbf{X}}^+)^{\top} \tilde{\mathbf{X}}^{\top} \hat{\boldsymbol{\epsilon}} = (\tilde{\mathbf{X}}^+)^{\top} \mathbf{E}^{\top} \hat{\boldsymbol{\epsilon}}$  as before. So

$$\tilde{\boldsymbol{\epsilon}} - \hat{\boldsymbol{\epsilon}} = (\mathbf{I} - \tilde{\mathbf{M}})(\mathbf{f} - \mathbf{E}\hat{\boldsymbol{\beta}}) - (\tilde{\mathbf{X}}^+)^{\top} \mathbf{E}^{\top} \hat{\boldsymbol{\epsilon}}.$$

The first term has norm at most  $\epsilon(\|\mathbf{y}\| + \sigma_1\|\hat{\boldsymbol{\beta}}\|) \leq \epsilon(1 + \kappa)\|\mathbf{y}\|$ , because  $\|\hat{\boldsymbol{\beta}}\| = \|\mathbf{X}^+\mathbf{y}\| \leq \|\mathbf{y}\|/\sigma_p$ . The second has norm at most  $\epsilon\sigma_1\|\hat{\boldsymbol{\epsilon}}\|/(\sigma_p(1 - \kappa\epsilon)) \leq \epsilon\kappa\|\mathbf{y}\|/(1 - \kappa\epsilon)$ . Adding and using  $1 + \kappa + \kappa/(1 - \kappa\epsilon) \leq 1 + 2\kappa/(1 - \kappa\epsilon)$  gives the bound Eq. 10.5.3. ■

The exact identity Eq. 10.5.4 shows where each term comes from. Perturbing the data along  $\mathcal{C}(\mathbf{X})$  moves the solution through  $\tilde{\mathbf{X}}^+$ , with amplification  $1/\sigma_p$ . This is the familiar  $\kappa$  of linear systems (Eq. 1.8.3). The second term is new. A perturbation  $\mathbf{E}$  that tilts a column of  $\mathbf{X}$  towards the residual direction changes the subspace, and the residual, however large, then “leaks” into the solution through  $(\tilde{\mathbf{X}}^\top \tilde{\mathbf{X}})^{-1}$ , with amplification  $1/\sigma_p^2$ . The residuals themselves, and hence the fitted values, are only  $\kappa$ -sensitive.

The  $\kappa^2$  term is not an artefact of the proof.

#### Example. (The squared condition number is attained)

Let  $\delta = 10^{-3}$  and

$$\mathbf{X} = \begin{pmatrix} 1 & 0 \\ 0 & \delta \\ 0 & 0 \end{pmatrix}, \quad \mathbf{y} = \begin{pmatrix} 0 \\ \delta \\ 1 \end{pmatrix},$$

so that  $\hat{\boldsymbol{\beta}} = (0, 1)^\top$ ,  $\hat{\boldsymbol{\epsilon}} = (0, 0, 1)^\top$ ,  $\kappa = 1000$  and  $\eta = 1$ . Perturb only the (3, 2) entry of  $\mathbf{X}$ , by  $\epsilon = 10^{-10}$ . This tilts the weak second column towards the residual. Its coefficient becomes  $(\delta^2 + \epsilon)/(\delta^2 + \epsilon^2)$ , and the relative change in  $\hat{\boldsymbol{\beta}}$  is  $1.00 \times 10^{-4}$ , equal to  $\kappa^2\epsilon = 1.00 \times 10^{-4}$ . A perturbation of relative size  $10^{-10}$  has changed the answer in the fourth digit.

```
import numpy as np
import statsmodels.api as sm

def ls(X, y):
    return np.linalg.lstsq(X, y, rcond=None)[0]

delta, eps = 1e-3, 1e-10
X = np.array([[1.0, 0.0], [0.0, delta], [0.0, 0.0]])
y = np.array([0.0, delta, 1.0])          # b = (0, 1), residual e = (0, 0, 1)
E = np.array([[0.0, 0.0], [0.0, 0.0], [0.0, eps]]) # ||E|| = eps ||X||
b = ls(X, y)
b_tilde = ls(X + E, y)
print("kappa =", np.linalg.cond(X))
```

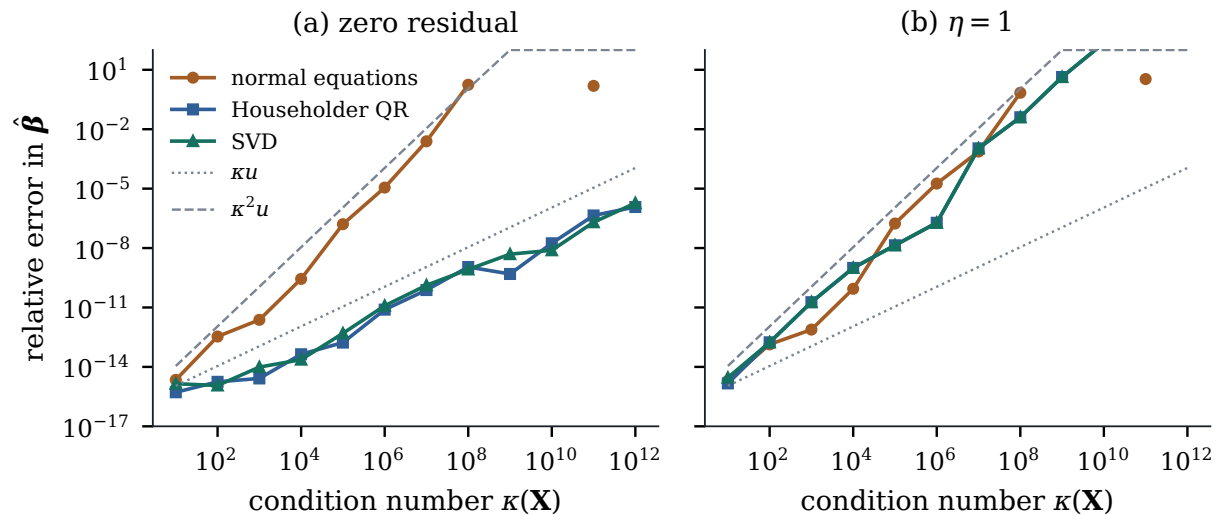


Figure 10.5.1: Relative error of the coefficients computed by the normal equations (Cholesky), Householder QR and the SVD, against  $\kappa(\mathbf{X})$ . (a) Consistent data,  $\hat{\mathbf{e}} = \mathbf{0}$ : QR and SVD track  $\kappa u$ , the normal equations track  $\kappa^2 u$ . (b) A large residual,  $\eta = 1$ : all three track  $\kappa^2 u$ , as [Theorem 10.5.1](#) predicts.

```
print("relative change in b:", np.linalg.norm(b_tilde - b) / np.linalg.norm(b))
print("kappa^2 * eps      :", np.linalg.cond(X) ** 2 * eps)
```

[Run this code online](#)

The script also draws 300 random problems with random perturbations and checks [Eq. 10.5.2](#) on each. The worst ratio of the actual change to the bound is 0.31.

## 10.5.2 What the theorem says about algorithms

A backward stable algorithm (Householder QR, Givens QR, modified Gram–Schmidt on the augmented matrix, the SVD) returns the exact solution for data perturbed as in [Eq. 10.5.1](#) with  $\epsilon \approx c u$ , where  $c$  grows modestly with the dimensions. Its relative error is therefore about

$$c u (\kappa + \kappa^2 \eta).$$

The normal equations incur a relative error of about  $c u \kappa^2$  *whatever*  $\eta$  is, because forming  $\mathbf{X}^\top \mathbf{X}$  already commits it ([Proposition 10.1.3](#)). Two regimes follow:

- **Small residual** ( $\eta \ll 1/\kappa$ ): QR gains a full factor  $\kappa$  over the normal equations. This is the regime of the experiment in [Section 6.10](#), where  $\mathbf{y} = \mathbf{X}\boldsymbol{\beta}$  exactly.
- **Large residual** ( $\eta$  of order one): every method, however stable, loses about  $2 \log_{10} \kappa$  digits, because the problem is that sensitive. The normal equations are then worse only by a modest factor.

[Figure 10.5.1](#) shows both regimes on test problems with prescribed condition numbers. With zero residual and  $\kappa = 10^6$ , the normal equations give relative error  $1.1 \times 10^{-5}$  and QR  $7.8 \times 10^{-12}$ . With  $\eta = 1$ , that is  $\|\hat{\mathbf{e}}\| = \|\mathbf{X}\|_2 \|\boldsymbol{\beta}\|$ , QR gives  $1.9 \times 10^{-7}$  and the SVD  $1.9 \times 10^{-7}$ . The  $\kappa^2$  term has caught up with the stable methods.

```
def by_cholesky(X, y):
```

```

L = np.linalg.cholesky(X.T @ X)
return np.linalg.solve(L.T, np.linalg.solve(L, X.T @ y))

def by_qr(X, y):
    Q, R = np.linalg.qr(X)
    return np.linalg.solve(R, Q.T @ y)

def by_svd(X, y):
    U, s, Vt = np.linalg.svd(X, full_matrices=False)
    return Vt.T @ ((U.T @ y) / s)

```

[Run this code online](#)

In regression  $\eta$  is rarely tiny, but QR keeps an advantage: its  $\kappa^2\eta$  term is a property of the problem, not of the method.

### 10.5.3 Column scaling

The normwise condition number depends on the units of the columns: measuring one regressor in millimetres instead of metres can multiply  $\kappa(\mathbf{X})$  by up to 1000 (Exercise (B2)), although the fit is unchanged. Equalizing the column lengths is essentially the best diagonal scaling.

#### Proposition 10.5.2 — Equilibrating the columns

Let  $\mathbf{X}$  have full column rank and let  $\mathbf{D}_0 = \text{diag}(1/\|\mathbf{x}_1\|, \dots, 1/\|\mathbf{x}_p\|)$ . For every nonsingular diagonal  $\mathbf{D}$ ,

$$\kappa(\mathbf{XD}_0) \leq \sqrt{p} \kappa(\mathbf{XD}).$$

*Proof.* The columns of  $\mathbf{XD}_0$  have unit length, so  $\sigma_1(\mathbf{XD}_0) \leq \|\mathbf{XD}_0\|_F = \sqrt{p}$ . For the smallest singular value, let  $\mathbf{z}$  be any nonzero vector and put  $\mathbf{w} = \mathbf{D}^{-1}\mathbf{D}_0\mathbf{z}$ . Then  $\|\mathbf{XD}_0\mathbf{z}\| = \|\mathbf{XD}\mathbf{w}\| \geq \sigma_p(\mathbf{XD})\|\mathbf{w}\|$  and  $\|\mathbf{w}\| \geq \min_j (1/(|d_j|\|\mathbf{x}_j\|))\|\mathbf{z}\|$ . The largest column length of  $\mathbf{XD}$ ,  $\max_j |d_j|\|\mathbf{x}_j\|$ , is at most  $\sigma_1(\mathbf{XD})$ , because the columns are images of unit vectors. So  $\sigma_p(\mathbf{XD}_0) \geq \sigma_p(\mathbf{XD})/\sigma_1(\mathbf{XD}) = 1/\kappa(\mathbf{XD})$ , and  $\kappa(\mathbf{XD}_0) \leq \sqrt{p} \kappa(\mathbf{XD})$ . ■

This is due to van der Sluis (1969). What makes it more than a remark about units is the form of the backward errors. Householder QR has a *columnwise* backward error (Theorem 10.3.2(c)). Rescaling the columns rescales the perturbations with them, so the computed solution is as accurate as if the columns had been equilibrated first. The rounding errors in  $\mathbf{X}^\top \mathbf{X}$  are entrywise small relative to  $|\mathbf{X}|^\top |\mathbf{X}|$  (Proposition 10.1.3), which has the same property. Both methods therefore behave according to the condition number of the *scaled* matrix,  $\kappa(\mathbf{XD}_0)$ , whatever the units. The rule of thumb should be applied to  $\kappa(\mathbf{XD}_0)$ , not to  $\kappa(\mathbf{X})$  as recorded.

#### Example. (Longley's data, exactly)

Longley (1967) built his macroeconomic data to test regression programs. With an intercept and six regressors,  $\kappa(\mathbf{X}) = 4.86 \times 10^9$  (Section 6.10; the data are described in Example (Macroeconomic series)). The data are published as decimals, so the exact

least squares solution can be computed in rational arithmetic. The script does so and counts the correct significant digits of the worst coefficient for each method:

| Method                                                    | Correct digits |
|-----------------------------------------------------------|----------------|
| Normal equations (Cholesky)                               | 7.2            |
| Householder QR                                            | 10.9           |
| SVD                                                       | 10.8           |
| statsmodels OLS                                           | 10.8           |
| Normal equations after scaling the columns to unit length | 7.0            |

The raw  $\kappa$  predicts about 6 digits from QR and none from the normal equations (Section 6.10). Both predictions are far too pessimistic. After scaling the columns to unit length,  $\kappa(\mathbf{XD}_0) = 4.33 \times 10^4$ . Then  $\kappa u \approx 5 \times 10^{-12}$  predicts about eleven digits from QR, and  $\kappa^2 u \approx 2 \times 10^{-7}$  about seven from the normal equations. That is what is observed. Scaling the columns explicitly before forming the normal equations changes nothing, as the argument above predicts. Centring as well reduces the condition number to  $1.11 \times 10^2$ . That is a reparameterization rather than a scaling (Theorem 6.7.1), and what remains is genuine collinearity among the regressors (Chapter 26).

```
lon = sm.datasets.longley.load_pandas()
XL = sm.add_constant(lon.exog.to_numpy())
yL = lon.endog.to_numpy()
names = ["const"] + list(lon.exog.columns)
print("kappa(X) as recorded:", f"{np.linalg.cond(XL):.3e}")
```

[Run this code online](#)

## 10.5.4 Numerical error and statistical error

The perturbation theorem applies to any perturbation of the data, including the rounding of recorded values, which is far above  $u$ . Longley's GNP deflator is given to one decimal place and the other series to whole units. Perturbing each regressor entry (the year is exact) uniformly within half a unit of its last recorded digit, 1000 times, the coefficients vary with standard deviations between 0.005 and 0.024 of their standard errors. The numerical error of the normal equations, the least accurate method, is at most  $4 \times 10^{-8}$  standard errors. Sampling error dominates, then comes the rounding of the published data, and the arithmetic is last by far. Beaton, Rubin and Barone (1976) made this point about Longley's data. A problem can be numerically delicate and statistically ordinary, or the reverse.

```
rng = np.random.default_rng(106)
half_unit = np.array([0.0, 0.05, 0.5, 0.5, 0.5, 0.5, 0.0]) # half a unit in the
# last digit
fitL = sm.OLS(yL, XL).fit() # year and constant are
# exact
reps = 1000
shifts = []
for _ in range(reps):
    Xp = XL + rng.uniform(-1, 1, size=XL.shape) * half_unit
    shifts.append(ls(Xp, yL) - fitL.params)
spread = np.std(shifts, axis=0)
ratio = spread / fitL.bse
```

```
for name, r in zip(names, ratio):
    print(f"{name:8s} sd of coefficient under rounding / standard error = {r:.2f}")
```

[Run this code online](#)

*Idea.* Least squares has a  $\kappa$  sensitivity, and a  $\kappa^2\eta$  one when the residual is large. Stable algorithms add nothing beyond this; the normal equations add  $\kappa^2$  regardless. Measure  $\kappa$  after equilibrating the columns.

## 10.5.5 Exercises

### 10.5.5.1 A. Check your understanding

#### Exercise A1.

In [Example \(The squared condition number is attained\)](#), find  $\tilde{\mathbf{b}}$  exactly and show that the relative change is  $\epsilon(1 - \epsilon)/(\delta^2 + \epsilon^2)$ . For which  $\epsilon$  is it close to  $\kappa^2\epsilon$ ?

### 10.5.5.2 B. Practice

#### Exercise B1.

Deduce from [Eq. 10.5.3](#) that the fitted values satisfy  $\|\tilde{\mathbf{X}}\tilde{\mathbf{b}} - \mathbf{X}\hat{\boldsymbol{\beta}}\| \leq \epsilon(2 + 2\kappa/(1 - \kappa\epsilon))\|\mathbf{y}\|$ . Why are fitted values never  $\kappa^2$ -sensitive, even though coefficients can be?

*Solution.*  $\tilde{\mathbf{X}}\tilde{\mathbf{b}} = \tilde{\mathbf{y}} - \tilde{\boldsymbol{\epsilon}}$  and  $\mathbf{X}\hat{\boldsymbol{\beta}} = \mathbf{y} - \hat{\boldsymbol{\epsilon}}$ , so the difference is  $\mathbf{f} - (\tilde{\boldsymbol{\epsilon}} - \hat{\boldsymbol{\epsilon}})$ , with norm at most  $\epsilon\|\mathbf{y}\| + \epsilon(1 + 2\kappa/(1 - \kappa\epsilon))\|\mathbf{y}\|$ . The  $\kappa^2$  term in the coefficients comes from  $(\tilde{\mathbf{X}}^\top \tilde{\mathbf{X}})^{-1} \mathbf{E}^\top \hat{\boldsymbol{\epsilon}}$ , a change of coordinates *within* the slightly rotated column space. Multiplying by  $\tilde{\mathbf{X}}$  turns the factor  $1/\sigma_p^2$  into  $1/\sigma_p$ . A direction that the columns barely span is a direction in which large coefficient changes produce only small changes in the fit.

#### Exercise B2.

Show that for consistent data ( $\hat{\boldsymbol{\epsilon}} = \mathbf{0}$ ) the bound reduces to  $2\kappa\epsilon/(1 - \kappa\epsilon)$ . Construct a perturbation  $\mathbf{f}$  of  $\mathbf{y}$  alone that changes  $\hat{\boldsymbol{\beta}}$  by a relative amount  $\kappa\epsilon$ , so that the bound is attained up to a factor of two.

### 10.5.5.3 C. Going deeper

#### Exercise C1.

Take data with  $\hat{\boldsymbol{\beta}} = \mathbf{v}_p$  and  $\hat{\boldsymbol{\epsilon}} \neq \mathbf{0}$ , and the perturbation  $\mathbf{E} = \epsilon\sigma_1 \hat{\boldsymbol{\epsilon}} \mathbf{v}_p^\top / \|\hat{\boldsymbol{\epsilon}}\|$ , with  $\mathbf{f} = \mathbf{0}$ . Use [Eq. 10.5.4](#) to show that, to first order in  $\epsilon$ ,  $\tilde{\mathbf{b}} - \hat{\boldsymbol{\beta}} = \epsilon\sigma_1 \|\hat{\boldsymbol{\epsilon}}\| \sigma_p^{-2} \mathbf{v}_p$ , so that the relative change is  $\epsilon\kappa^2\eta$ . Compare with the first-order bound  $\epsilon(2\kappa + \kappa(\kappa + 1)\eta)$ .